NOTE: Logical Data Independence is more difficult to achieve.

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39) What are the three levels of data abstraction?

Following are three levels of data abstraction:

Physical level: It is the lowest level of abstraction. It describes how data are stored.

Logical level: It is the next higher level of abstraction. It describes what data are stored in the database and what relationship among those data.

View level: It is the highest level of data abstraction. It describes only part of the entire database.

For example- User interact with the system using the GUI and fill the required details, but the user doesn't have any idea how the data is being used. So, the abstraction level is absolutely high in VIEW LEVEL.

Then, the next level is for PROGRAMMERS as in this level the fields and records are visible and the programmer has the knowledge of this layer. So, the level of abstraction here is a little low in VIEW LEVEL.

And lastly, physical level in which storage blocks are described.

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40) What is Join?

The Join operation is one of the most useful activities in relational algebra. It is most commonly used way to combine information from two or more relations. A Join is always performed on the basis of the same or related column. Most complex queries of SQL involve JOIN command.

There are following types of join:

- Inner joins: Inner join is of 3 categories. They are:
    - Theta join